

DEEDY: A Thai-Centric Multi-Agent Framework for Social-Media Marketing Campaign Analysis

Sawit Koseeyaumporn

Department of Computer Engineering

King Mongkut's University of Technology Thonburi, Bangkok, Thailand

sawit.kosee@kmutt.ac.th

Abstract—This paper presents DEEDY (Digital Environment for Emergent Dynamics and Yield), a Thai-centric multi-agent framework for analyzing social-media campaign response. DEEDY separates factual campaign evidence from authentic consumer reactions, constructs a grounded simulated world, and combines language-model agents with a scalable stochastic population. The evaluation uses 5,000 comments from five Thai campaigns. Across five experiments, 1,300 generation calls test grounding, Thai prompting, response models, social mechanisms, and message clarity. A separate GLM-5.3-Flash judge evaluates 741 real-synthetic comment pairs twice with reversed presentation order. Authentic comments were preferred in 59.3% of judgments, although synthetic reactions received higher naturalness and campaign-relevance scores. The system therefore demonstrates an auditable workflow and useful sensitivity analysis, but not behavioral fidelity, campaign effectiveness, or deterministic forecasting.

Index Terms—LLM agents, social-media marketing, campaign analysis, social simulation, Thai NLP, agent-based modeling, MiroFish

I. INTRODUCTION

Social-media campaigns develop through exposure, reply, sharing, and recommendation rather than through isolated comments. Sentiment counts describe a snapshot but not the interaction process that produces it. Agent-based modeling (ABM) provides a bottom-up representation of local decisions and their collective effects [1]; LLM agents add context-dependent language, memory, and open-ended actions [2], [3].

DEEDY adapts the MiroFish/OASIS simulation workflow [4], [5] to campaign analysis. It builds audience actors from approved campaign evidence, simulates exposure and social actions, and compares the resulting discourse with authentic reactions that were hidden during generation. Counterfactual differences are treated as hypotheses, not predictions of a specific person or the Thai market.

Thai campaigns require more than prompt translation. Informal comments mix Thai and English, emoji, non-standard spelling, sarcasm, indirect disagreement, and culturally specific politeness. Thai resources have improved [6], [7], [8], [9], but linguistic form does not guarantee cultural competence [10]. Moreover, plausible generated text does not prove behavioral fidelity [11], [12], [13]. DEEDY therefore separates construction evidence from evaluation reactions and reports uncertainty across repeated conditions.

Our contributions are threefold:

- a provenance-preserving Thai campaign corpus and evidence split;
- a grounded hybrid simulation with configurable exposure, memory, and Thai-language action generation; and
- a leakage-resistant evaluation combining distributional metrics, reversed-order LLM judging, and planned native-Thai review.

II. RELATED WORK

Generative Agents introduced memory, reflection, and planning for believable behavior [2]; S^3 , OASIS, and GenSim extended this idea to networked social environments, multiple online actions, and reusable simulations [14], [5], [15]. MiroFish connects a reality seed to graph-grounded personas, social simulation, and report generation [4]. DEEDY adapts this workflow to Thai campaign analysis and adds an explicit separation between evidence used to construct a scenario and evidence used to evaluate it.

This separation is important because plausible language is not equivalent to behavioral validity. Algorithmic fidelity compares generated and human population distributions [11], while conditioned-comment prediction directly compares generated and authentic responses to the same stimulus [13]. Generative ABMs should therefore be validated incrementally before counterfactual claims are made [16].

Thai social-media analysis adds informal spelling, code-mixing, emoji, sarcasm, indirect stance, and culturally specific politeness. Thai NLP tools and pretrained models provide useful linguistic infrastructure [6], [7], [8], [9], but cultural knowledge remains a separate capability [10]. Conversation context is also necessary because sarcasm can reverse literal polarity [17]. These findings motivate evaluation of naturalness, cultural fit, stance, and authenticity rather than sentiment alone.

III. DEEDY SYSTEM ARCHITECTURE AND WORKFLOW

A. Evidence Preparation

Public campaign posts and comments are collected from Facebook, Instagram, and TikTok through recorded Apify queries [18]. Text is preserved with emoji, Thai-English code-mixing, repeated characters, and thread cues because these may carry pragmatic meaning. Direct identifiers are removed from research views, while campaign, platform, post, time, and collection provenance are retained. The resulting corpus is not treated as a random sample of Thai consumers: platform

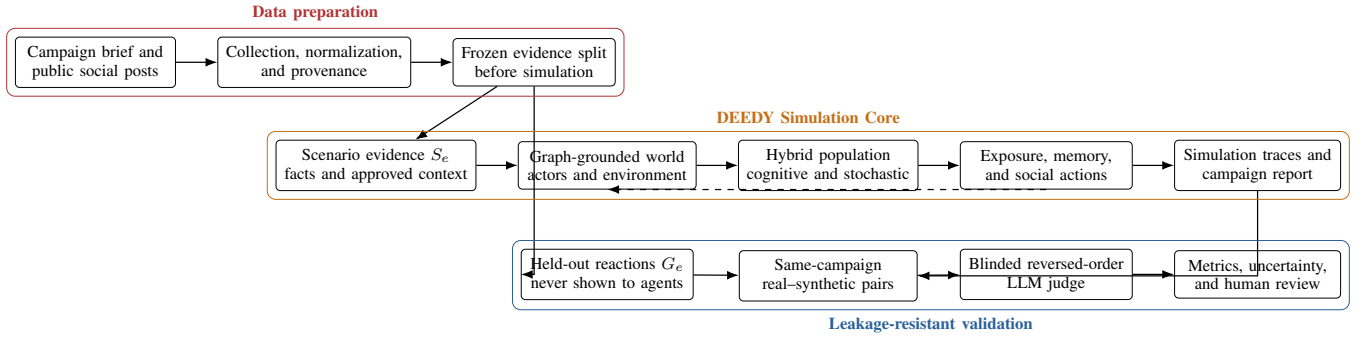


Fig. 1. DEEDY architecture and executed evaluation workflow. Campaign facts construct the simulated world, while authentic reactions remain hidden until outputs are scored. The dashed arrow represents memory feedback within the simulation.

visibility, ranking, moderation, deletion, and source selection can affect which comments are observable.

For campaign e , evidence is partitioned before generation as

$$D_e = S_e \dot{\cup} G_e, \quad (1)$$

where S_e contains only facts and context permitted to construct the scenario, and G_e contains authentic reactions reserved for evaluation. Comments, their summaries, and derived sentiment proportions from G_e are not supplied to agents. This design prevents the simulator from repeating the answers against which it will later be scored.

B. DEEDY Simulation Core

Figure 1 presents the conceptual architecture. The deployed component is the *DEEDY Simulation Core*; MiroFish supplies the world-construction and social-simulation foundation [4]. The core executes five steps: (1) scenario construction, (2) actor and environment initialization, (3) hybrid-agent decision making, (4) exposure and network propagation, and (5) trace aggregation and report generation.

1) *Scenario, Actors, and Memory*: Campaign facts are organized as entities, relations, and retrievable scenario memories. This graph-grounding step is retained because an agent must recover the correct product, campaign condition, creator, and prior event when deciding how to react; it is not used to inject authentic evaluation comments. Audience roles are sampled with interests, prior attitude, activity, language register, Thai-English mixing, and uncertainty. Evidence-backed attributes remain separate from sampled attributes, and unsupported details remain assumptions. The environment defines the platform action space, time step, feed capacity, and exposure policy.

Each cognitive agent maintains short-term interaction memory and longer-term scenario memory. At a simulation step, the agent retrieves relevant facts, observes a bounded feed and recent interaction context, updates its private stance, and selects like, share, comment, reply, or silence. Thai generation preserves code-mixing and uncertainty without forcing dialect caricatures. Thread context is available for replies because Thai stance and sarcasm may depend on earlier turns [17].

2) *Hybrid Population and Propagation*: The population combines a smaller cognitive tier with a larger stochastic tier. Cognitive agents retrieve relevant memories, observe a bounded feed, update their private stance, and generate a Thai action or remain silent. Stochastic agents preserve population scale using a probabilistic response conditioned on prior attitude, exposure, neighbors, and the preceding social signal. For an exposed stochastic agent i at time t ,

$$\Pr(a_{i,t} \neq \emptyset) = \sigma(\beta_0 + \beta_b b_i + \beta_d^\top \mathbf{d}_i + \beta_s s_{t-1} + \beta_x x_{i,t}). \quad (2)$$

The action space includes like, share, comment, reply, and silence. Initial exposure can be chronological, interest-based, popularity-based, or seeded through influential agents; later exposure propagates through the configured network. These mechanisms are experimental assumptions, not reconstructions of a platform’s proprietary recommender.

3) *Outputs and Validation Path*: Every output connects an action to its actor, exposure, retrieved evidence, parent action, and simulated time. The simulation report summarizes sentiment, stance, narratives, reach, engagement, influential actors, and disagreement across repeated runs. Separately, visible generated reactions are paired with same-campaign authentic comments from G_e for blinded evaluation. Aggregate reference distributions are loaded only after generation and judging. This traceable separation supports error analysis and counterfactual comparison but does not make a simulated difference a real-world causal effect.

IV. EVALUATION AND EXPERIMENTAL RESULTS

A. Data and Experimental Design

The evaluation covers five Thai campaigns: Parameter Gelato, KFC Bucket Ware, MK Buffet, Walk/Run for Glasses by Top Charoen, and Thai Chuai Thai Plus. The corpus contains 5,000 authentic comments from 33 posts, with 1,000 comments per campaign. Aggregate sentiment analyses were used only after generation. The available crawl contained no reply links, so observed cascade correspondence could not be evaluated; network results are reported only as model sensitivity.

Experiments 1–3 test grounding, Thai prompting, and the response model. Experiment 4 varies feed and network assumptions and generates agent reactions. Experiment 5 compares a

baseline campaign packet with a clarity frame that makes conditions and uncertainties more explicit. Ten random seeds are used per campaign-condition combination. Jensen–Shannon distance (JSD) measures sentiment-distribution divergence, while text quality is assessed with diversity, semantic correspondence [19], and blinded judgment. Tables I and II report the principal numeric results.

Across Experiments 1–5, 1,300 generation calls completed. Experiments 4–5 produced 6,400 agent records from 800 calls. Of these, 741 calls contained at least one visible reaction; 59 all-silence calls were retained in action-rate analysis but excluded from text comparison. Experiment 1 did not establish a grounding advantage, Experiment 2 did not establish a Thai-context advantage, and Experiment 3 did not establish superiority of either LLM or the stochastic baseline. High Thai-character ratios were not interpreted as naturalness or cultural competence.

Experiment 4 changed LLM outputs only modestly across prompt-level feed/network descriptions and under-produced negative reactions. In the separate non-generative mechanism sweep, the matched-random graph produced action rates of 0.968–0.970 and reach of 0.997, whereas the post-affiliation proxy produced action rates of 0.400–0.402 and reach of 0.461–0.463. Network assumptions therefore dominated population-size changes. Because the corpus did not preserve reply edges, neither topology represents an observed Thai interaction network.

For Experiment 5, clarity improved DeepSeek JSD by 0.0075 but worsened Qwen JSD by 0.0014. Across 100 matched generator–campaign–seed pairs, the clarity-minus-baseline change was -0.0030 (95% empirical interval -0.1888 – 0.1261). The interval spans zero and does not establish a message-clarity advantage.

B. Blinded LLM-as-a-Judge Evaluation

Each eligible synthetic reaction was paired with a distinct authentic comment from the same campaign. Direct identifiers and contact patterns were removed; the judge received only the campaign facts and two unlabeled texts. Each of the 741 pairs was evaluated twice with A/B order reversed, producing 1,482 valid judgments. GLM-5.3-Flash scored Thai naturalness, campaign relevance, social-media plausibility, pragmatic/cultural fit, contextual specificity, and unsupported-claim risk on five-point scales. It also predicted sentiment and selected which text appeared more authentic. Pair reversal was used because LLM judges may exhibit presentation and position biases [20]. Aggregate sentiment references were hidden until all judgments were complete.

The authentic comment was preferred in 59.3% of judgments, the synthetic reaction in 40.6%, and a tie in 0.1%. Treating a tie as half a win gives a synthetic-realism score of 0.406 (bootstrap 95% CI 0.374–0.438), below the 0.5 indifference point. Reversed-order preference agreed for 83.5% of pairs, while candidate A was selected in 55.9% of non-ties, confirming measurable position sensitivity.

Table III reveals a useful tension. Synthetic reactions were rated more natural, relevant, culturally appropriate, and specific, yet authentic comments were more often selected as real. Social plausibility and unsupported-claim risk showed no clear difference. The judge-labeled sentiment distribution diverged less from the aggregate reference for authentic comments (JSD = 0.077) than for synthetic reactions (0.215), reinforcing the under-production of negative response. KFC was the only campaign with a synthetic-realism score slightly above 0.5, whereas Top Charoen had the largest authenticity gap. Campaign slices are diagnostic because sample eligibility and judge priors may differ.

C. Campaign-Level Results

Table IV reports all five campaign slices. No campaign’s realism interval lay wholly above the 0.5 indifference point. KFC was closest to indifference, while Top Charoen was most readily distinguished from authentic discussion. Sentiment correspondence did not always move with authenticity: Top Charoen’s synthetic sentiment distribution was close to the aggregate reference even though its generated comments were rarely preferred as real. Distribution matching and comment-level realism therefore measure different properties and should be reported together.

V. DISCUSSION, LIMITATIONS, AND CONCLUSION

The results do not simply make DEEDY appear more accurate. They expose a "too polished and too relevant" failure mode: prompted agents produce coherent standalone responses, whereas authentic threads contain fragments, tagging, repetition, jokes, and side conversations. Optimizing only rubric scores could therefore reduce realism. Experiment 4 evaluates sensitivity to specified feed and network conditions, not reconstruction of platform exposure. Experiment 5 evaluates reactions to hypothetical framing, not a causal effect on consumers. The simulations are consequently most useful for scenario exploration and failure discovery.

The study has four principal limitations. First, manually selected public posts form a convenience sample rather than a representative Thai population. Second, missing reply edges prevent validation of propagation and cascade structure. Third, the sentiment reference is an aggregate weak annotation, not an independent human gold standard. Fourth, a single external LLM judge can share model priors with the generators and showed position sensitivity. Sanitization reduces but cannot eliminate privacy risk when authentic text is sent to an external service. Future evaluation should use a pre-registered campaign-level holdout, reply- and time-preserving traces, a Thai-specialized generator, multiple judges, and blinded native-Thai raters with reported agreement.

DEEDY contributes a leakage-resistant architecture for connecting grounded Thai campaign scenarios, hybrid multi-agent simulation, and authentic-response evaluation. The completed pilot demonstrates end-to-end execution and reveals where the model departs from observed discourse; it does not establish market forecasting, campaign effectiveness, or behavioral

TABLE I

EXPERIMENTS 1–3: MEAN SENTIMENT JSD OVER FIVE CAMPAIGNS AND TEN SEEDS. THE AGGREGATE REFERENCE IS A DIAGNOSTIC CALIBRATION TARGET, NOT AN INDEPENDENT HOLDOUT. NO REPORTED PAIRED INTERVAL EXCLUDED ZERO.

Experiment	Condition	DeepSeek	Qwen	Stochastic ABM
Exp. 1	Grounded scenario	0.2572	0.2859	–
Exp. 1	Ungrounded prompt	0.2635	0.2388	–
Exp. 2	Thai-context prompt	0.2572	0.2859	–
Exp. 2	Translated/unadapted	0.2604	0.2523	–
Exp. 2	Normalized, no context	0.2460	0.2548	–
Exp. 3	Response-model comparison	0.2572	0.2859	0.2779

TABLE II

EXPERIMENTS 4–5 LLM REACTION RESULTS. EXPERIMENT 4 ENTRIES ARE RANGES OVER SIX FEED/NETWORK DESCRIPTIONS; EXPERIMENT 5 ENTRIES AVERAGE FIVE CAMPAIGNS AND TEN SEEDS. VISIBLE ACTION EXCLUDES SILENCE.

Experiment	Generator	Condition	Runs	Sentiment JSD	Visible action	Negative share
Exp. 4	DeepSeek	Feed/network range	300	0.2995–0.3233	0.760–0.783	0.008–0.023
Exp. 4	Qwen	Feed/network range	300	0.3274–0.3432	0.740–0.753	0.000–0.005
Exp. 5	DeepSeek	Baseline	50	0.3189	0.7775	0.0100
Exp. 5	DeepSeek	Clarity	50	0.3114	0.8200	0.0250
Exp. 5	Qwen	Baseline	50	0.3335	0.7450	0.0000
Exp. 5	Qwen	Clarity	50	0.3349	0.7500	0.0050

TABLE III

BLINDED JUDGE RESULTS OVER 741 REAL–SYNTHETIC PAIRS. SCORE GAPS ARE SYNTHETIC MINUS REAL; RISK IS THE ONLY DIMENSION WHERE HIGHER IS WORSE.

Dimension	Real	Synthetic	Gap (95% CI)
Thai naturalness	3.77	4.19	0.42 [0.29, 0.57]
Campaign relevance	2.83	4.46	1.64 [1.51, 1.77]
Social plausibility	3.84	3.88	0.04 [−0.11, 0.20]
Cultural fit	3.57	4.07	0.50 [0.35, 0.65]
Context specificity	2.23	3.14	0.91 [0.76, 1.05]
Unsupported-claim risk	1.83	1.87	0.04 [−0.06, 0.14]

TABLE IV

CAMPAIGN-LEVEL JUDGE RESULTS. REALISM EQUALS SYNTHETIC PREFERENCE PLUS HALF THE TIE SHARE. JSD IS SHOWN AS AUTHENTIC/SYNTHETIC; LOWER IS BETTER.

Campaign	Pairs	Realism (95% CI)	JSD (R/S)
Parameter Gelato	155	0.431 [0.363, 0.498]	0.181/0.428
KFC Bucket Ware	142	0.511 [0.440, 0.585]	0.288/0.274
MK Buffet	149	0.450 [0.376, 0.527]	0.123/0.220
Top Charoen	155	0.247 [0.182, 0.313]	0.068/0.049
Thai Chuai Thai Plus	140	0.404 [0.332, 0.475]	0.174/0.376

fidelity. This distinction is central to using generative simulation as an auditable hypothesis tool rather than a substitute for field research.

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